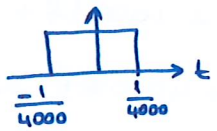
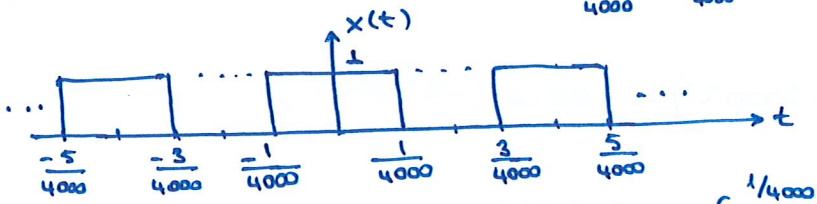
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ELEC 361 Analog Communication Systems Midterm Exam **Fall 2023**
Prof. Dr. Oğuz Kucur **Time: 100 min.** **1.12.2023**

1)(30 p.) a) Find exponential Fourier series expansion of the signal $x(t) = \sum_{k=-\infty}^{\infty} \Pi(2000t - 2k)$. Find its power spectral density (PSD) function and sketch for $|f| \leq 3.5\ kHz$.
b) The signal $x(t)$ is applied to the input of the linear time invariant (LTI) system with the impulse response $h(t) = 20000Sinc(10000t - 100000)$. Find the transfer function of the system and sketch it. Determine if the system is distortionless or not for $x(t)$. State reasoning.
c) For the input signal $x(t) = 20Cos(4000\pi t) + 10Cos(8000\pi t)$, state if the system in part b) is distortionless or not. State reasoning. Find the PSD function, autocorrelation function and power of the output signal $y(t)$.

$$x(t) = \sum_{k=-\infty}^{\infty} \Pi\left(\frac{t - k/1000}{1/2000}\right) \qquad \Pi\left(\frac{t}{1/2000}\right) \Rightarrow$$





$T_0 = \frac{1}{4000} = 10^{-3}\ sn$

$f_0 = 1\ kHz$, $\omega_0 = 2000\pi$

a) $x(t) = \sum_{n=-\infty}^{\infty} x_n . e^{jn\omega_0 t}$, $x_0 = \frac{1}{T_0}\int_{T_0} x(t)dt = 1000\int_{-\frac{1}{4000}}^{1/4000} 1\ dt = 1000\frac{2}{4000} = \frac{1}{2}$

$x_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t} dt = 10^3 \int_{-\frac{1}{4}.10^{-3}}^{\frac{1}{4}.10^{-3}} 1 . e^{-jn2000\pi t} dt = \frac{-1}{jn.2\pi 10^3}\left\{ e^{-jn.2.10^3 \frac{1}{4}10^{-3}\pi} - e^{jn.2.10^3\frac{1}{4}10^{-3}\pi} \right\}$

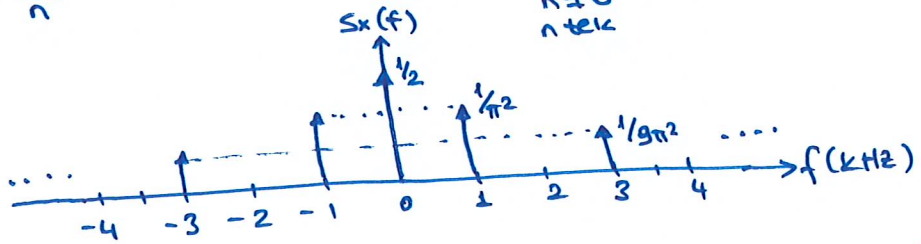
$= \frac{1}{\pi n}.\frac{1}{2j}\left\{ e^{j\frac{n}{2}\pi} - e^{-j\frac{n}{2}\pi} \right\}$

$= \frac{1}{\pi n}.\sin\left(\frac{\pi n}{2}\right)$, $n \neq 0$

$x(t) = \frac{1}{2} + \sum_{\substack{n\neq 0 \\ n\ tek}} \frac{1}{\pi n}\sin\left(\frac{\pi n}{2}\right) e^{jn2000\pi t}$

$\hookrightarrow \sin\left(\frac{\pi n}{2}\right) = \begin{cases} 1 & , n = 1,5,9,13.. \\ -1 & , n = 3,7,11,.. \end{cases}$

$S_x(f) = \sum_n |x_n|^2 \delta(f - nf_0) = \frac{1}{2}\delta(f) + \sum_{\substack{n\neq 0 \\ n\ tek}} \frac{1}{\pi^2 n^2}.\delta(f - n1000)$



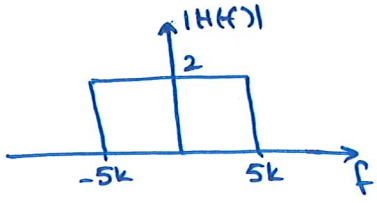
b) $x(t) \rightarrow [h(t)] \rightarrow y(t)$

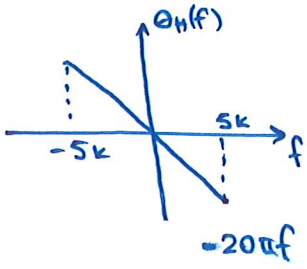
$h(t) = 20000\,sinc(\underbrace{10000t - 100000}_{10000(t-10)}) \longrightarrow H(f) = 2.\Pi\left(\frac{f}{10000}\right)e^{-j2\pi f 10}$

$\tau = 10000$, $A = 2$, $t_0 = 10$

$|H(f)| = 2\Pi\left(\frac{f}{10000}\right)$

$\theta_H(f) = -2\pi 10 f$



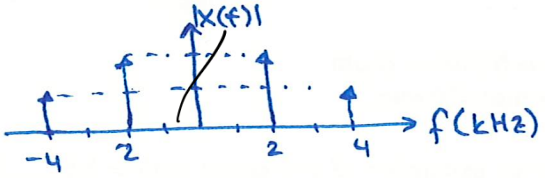


$\Rightarrow |H(f)|$, $x(t)$'nin spektrumunun tamaminda $|K| \neq 0$ olmadigindan bu işaret için bozulmasız değildir.

c) $x(t) = 20\cos(2\pi 2000t) + 10\cos(2\pi 4000t)$

$\downarrow$

$X(f) = 10\delta(f-2000) + 10\delta(f+2000) + 5\delta(f-4000) + 5\delta(f+4000)$



$|H(f)|$, $x(f)$'in BG'si içinde sabit, $\theta_H(f)$'de lineer olduğu için sistem bu işaret için bozulmasızdır.

$K=2$, $t_d=10$



$$S_y(f) = |H(f)|^2 . S_x(f)$$

$$S_x(f) = 100\delta(f-2000) + 100\delta(f+2000) + 25\delta(f-4000) + 25\delta(f+4000)$$

$$S_y(f) = 400\delta(f-2000) + 400\delta(f+2000) + 100\delta(f-4000) + 100\delta(f+4000)$$

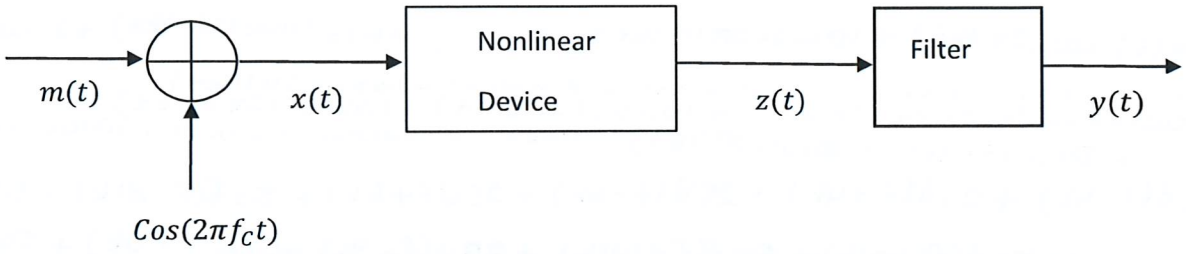
$\mathcal{F}^{-1}\downarrow$

$$R_y(\tau) = 800\cos(2\pi 2000\tau) + 200\cos(2\pi 4000\tau)$$

1.yol: $P_y = R_y(0) = 800 + 200 = 1000$ W

2.yol: $P_y = \int_{-\infty}^{\infty} S_y(f)\,df = 2.400 + 2.100 = 1000$ W

2.(40 p.) a) The message signal $m(t)$ with bandwidth (BW) B Hz is applied to the system consisting of an adder, a nonlinear device and a filter, respectively shown in the figure below. The adder adds $m(t)$ and $Cos(2\pi f_C t)$. The nonlinear device has the input-output relationship $z(t) = 5x^2(t)$. Find the output signal $y(t)$ for each of the following cases: The filter is i) a low pass filter (LPF) with a gain of 5, BW=$2B$, ii) a bandpass filter (BPF) with a gain of 2, BW=$2B$, center frequency f_C.



b) For the case ii), what kind of modulation signal is obtained? State reasoning. If $m(t) = 10Cos(10000\pi t) + 20Cos(20000\pi t)$ and $f_C = 50\ kHz$, express $y(t)$. Find its spectra and sketch it. Find its power and bandwidth. Describe and find the efficieny of the modulation. c) The demodulator consists of a multiplier using the estimated carrier signal $Cos(102000\pi t)$ and a filter. Determine the specifications of the filter. At each step of the demodulator, find the signals in both time and frequency domains and sketch the spectra. Comment on the signal at the output of the demodulator.

a) $x(t) = m(t) + \cos(2\pi f_c t) \xrightarrow{\mathcal{F}} X(f) = M(f) + \frac{1}{2}\delta(f-f_c) + \frac{1}{2}\delta(f+f_c)$

$z(t) = 5x^2(t) = 5\left[m(t) + \cos(2\pi f_c t)\right]^2 = 5\left[m^2(t) + 2m(t)\cos(2\pi f_c t) + \underbrace{\cos^2(2\pi f_c t)}_{\frac{1}{2}+\frac{1}{2}\cos(2\pi 2f_c t)}\right]$

$z(t) = 5m^2(t) + 10m(t)\cos(2\pi f_c t) + \frac{5}{2} + \frac{5}{2}\cdot\cos(2\pi 2f_c t)$

$\mathcal{F}\downarrow$

$Z(f) = \underbrace{5M(f) * M(f)}_{BG\text{'si } 2B} + 5M(f-f_c) + 5M(f+f_c) + \frac{5}{2}\delta(f) + \frac{5}{4}\delta(f-2f_c) + \frac{5}{4}\delta(f+2f_c)$

i) LPF, Gain=5, BW=2B $\quad Y(f) = 25M(f) * M(f) + \frac{25}{2}\delta(f) \longrightarrow y(t) = 25m^2(t) + \frac{25}{2}$

ii) BPF, Gain=2, BW=2B, MF=f_c $\quad Y(f) = 10M(f-f_c) + 10M(f+f_c) \longrightarrow y(t) = 20m(t)\cos(2\pi f_c t)$

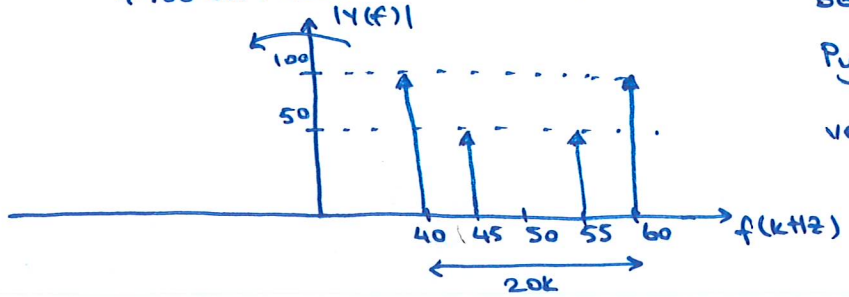
b) → DSB-SC-AM, LC yok.

$m(t) = 10\cos(2\pi 5kt) + 20\cos(2\pi 10kt)$, $f_c = 50kHz$

$y(t) = 200\cdot\cos(2\pi 5kt)\cdot\cos(2\pi 50kt) + 400\cdot\cos(2\pi 10kt)\cdot\cos(2\pi 50kt)$

$= 100\cos(2\pi 55kt) + 100\cos(2\pi 45kt) + 200\cos(2\pi 60kt) + 200\cos(2\pi 40kt)$

$Y(f) = 50\delta(f-55k) + 50\delta(f+55k) + 50\delta(f-45k) + 50\delta(f+45k) + 100\delta(f-60k) + 100\delta(f+60k)$
$+ 100\delta(f+40k) + 100\delta(f-40k)$

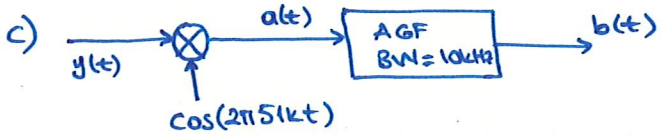


$BG\{y(t)\} = 20kHz$

$P_y = 4.100^2 + 4.50^2 = 50kW$

verim $= \frac{P_m}{P_y} = \frac{\frac{10^2}{2} + \frac{20^2}{2}}{50k} = 0{,}005$

c)



$a(t) = y(t)\cos(2\pi 51kt) = 20\,m(t)\underbrace{\cos(2\pi 50k).\cos(2\pi 51kt)}$

$\frac{1}{2}\cos(2\pi 1kt) + \frac{1}{2}\cos(2\pi 101kt)$

$= 10\,m(t)\cos(2\pi 1kt) + 10\,m(t)\cos(2\pi 101kt)$, $m(t) = 10\cos(2\pi 5kt) + 20\cos(2\pi 10kt)$

$= 50\cos(2\pi 6kt) + 50\cos(2\pi 4kt) + 100\cos(2\pi 9kt) + 100\cos(2\pi 11kt)$

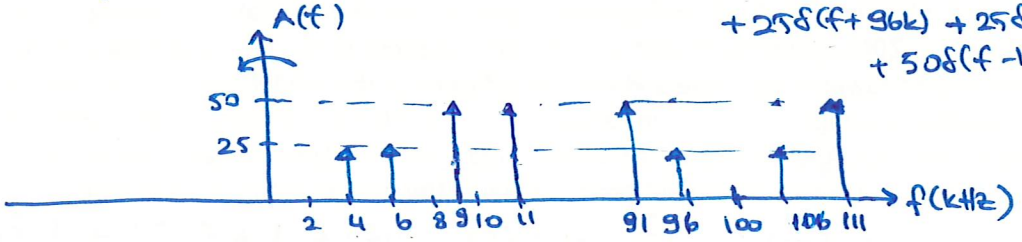
$+ 50\cos(2\pi 96kt) + 50\cos(2\pi 106kt) + 100\cos(2\pi 91kt) + 100\cos(2\pi 111kt)$

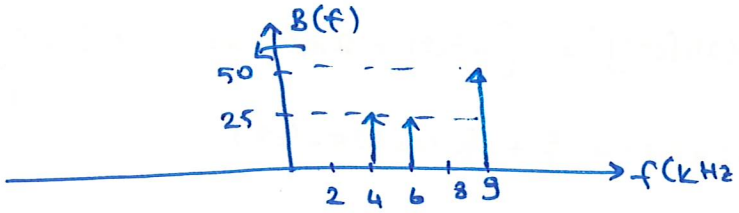
$A(f) = 25\delta(f-4k) + 25\delta(f+4k) + 25\delta(f-6k) + 25\delta(f+6k) + 50\delta(f-91k) + 50\delta(f+91k)$

$+ 50\delta(f-111k) + 50\delta(f+111k) + 50\delta(f-9k) + 50\delta(f+9k) + 25\delta(f-96k)$

$+ 25\delta(f+96k) + 25\delta(f-106k) + 25\delta(f+106k)$

$+ 50\delta(f-11k) + 50\delta(f+11k)$





$B(f) = 50\delta(f-9k) + 50\delta(f+9k) + 25\delta(f-4k) + 25\delta(f+4k) + 25\delta(f-6k) + 25\delta(f+6k)$

$b(t) = 100\cos(2\pi 9kt) + 50\cos(2\pi 4kt) + 50\cos(2\pi 6kt)$

* taşıyıcı frekans doğru şekilde tahmin edilmediğinden mesaj işareti geri elde edilememiştir.

(30 p.) The message signal $m(t) = 5 - 10Sin(2000\pi t)$ is used to modulate the carrier signal to obtain Double Sideband (DSB) Large Carrier (LC) Amplitude Modulation (AM) signal.

a) If the efficiency is required to be 15 %, find the modulation index.

b) If the carrier signal is $c(t) = 20Cos(20000\pi t)$, find the powers of the carrier, the sidebands and the modulated signal.

c) Find the modulated signal and sketch it.

d) Find the spectra of the normalized message signal and modulated signal. Sketch them.

Hint: $Sin(x).Cos(y) = \frac{1}{2}[Sin(x-y) + Sin(x+y)]$

a) $\text{verim} = \dfrac{\mu^2 P_{mn}}{1+\mu^2 P_{mn}} = 0.15$

$m(t) = 5 - 10\sin(2000\pi t) \longrightarrow m_p = 15 \Rightarrow m_n(t) = \frac{1}{3} - \frac{2}{3}\sin(2000\pi t)$

$P_{mn} = \left(\frac{1}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 \cdot \frac{1}{2} = \frac{3}{9} = \frac{1}{3}\,W$

$\mu^2 \cdot \frac{1}{3} = \frac{15}{100} + \mu^2 \cdot \frac{1}{3} \cdot \frac{15}{100} \longrightarrow \mu^2 = 0{,}53 \rightarrow \mu = 0{,}7276 \cong 0{,}728$

b) $c(t) = 20\cos(2\pi\, 10k\, t) \longrightarrow A_c = 20$, $f_c = 10\,kHz$

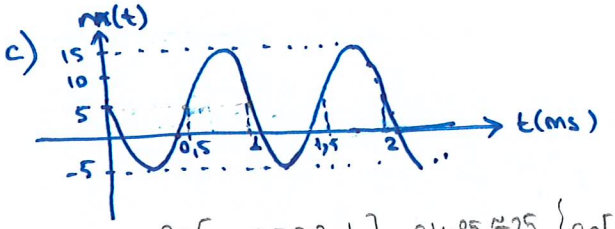
$y(t) = A_c[1 + \mu m_n(t)] \cdot \cos(2\pi f_c t) = 20\left[1 + 0{,}24 - 0{,}48\sin(2\pi 1000 t)\right]\cos(2\pi 10k t)$

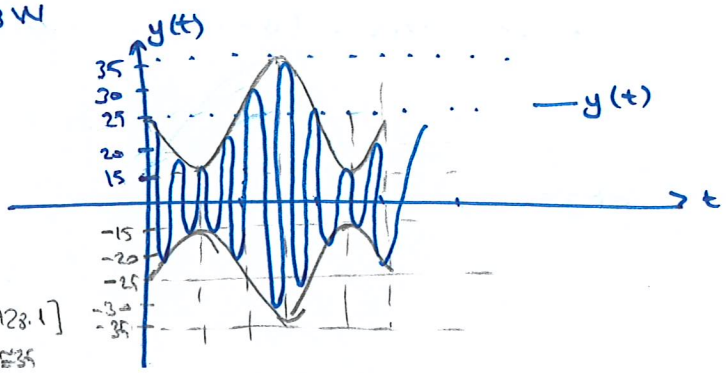
$P_c = \frac{A_c^2}{2} = \frac{20^2}{2} = 200\,W$

$P_s = \mu^2 \frac{A_c^2}{2} P_{mn} = (0{,}728)^2 \cdot \frac{20^2}{2} \cdot \frac{1}{3} = 35{,}33\,W$

$P_y = P_c + P_s = 235{,}33\,W$

c)





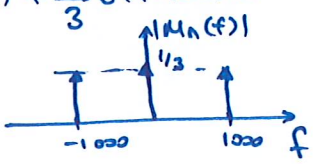
$20\left[1 + 0{,}728\cdot\frac{1}{3}\right] = 24{,}85 \cong 25$

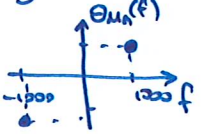
$20\left[1 + 0{,}728\cdot\frac{-1}{3}\right] = 15{,}14 \cong 15$

$20[1 + 0{,}728 \cdot 1] = 34{,}56 \cong 35$

d) $m_n(t) = \frac{1}{3} - \frac{2}{3}\sin(2\pi 1000 t) \longrightarrow M_n(f) = \frac{1}{3}\delta(f) - \frac{2}{3}\cdot\frac{1}{2j}\left[\delta(f-1000) - \delta(f+1000)\right]$

$= \frac{1}{3}\delta(f) + \frac{1}{3}\delta(f-1000).e^{j\pi/2} + \frac{1}{3}\delta(f+1000)e^{-j\pi/2}$

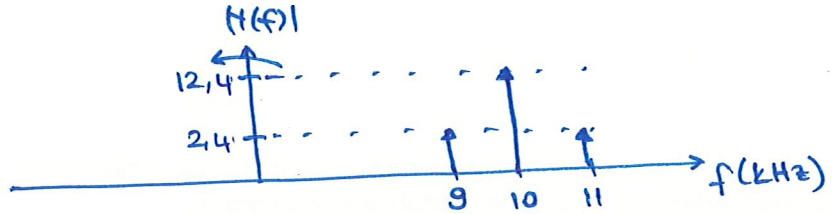


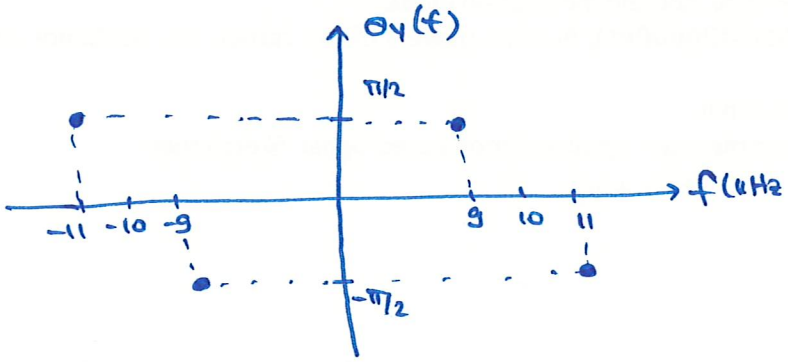


$y(t) = 24{,}8\cos(2\pi 10k t) - 9{,}6\underbrace{\sin(2\pi 1000 t).\cos(2\pi 10k t)}_{\frac{1}{2}\sin(-2\pi 9k t) + \frac{1}{2}\sin(2\pi 11k t)}$

$= 24{,}8\cos(2\pi 10k t) + 4{,}8\sin(2\pi 9k t) + 4{,}8\sin(2\pi 11k t)$

$Y(f) = 12{,}4\delta(f-10k) + 12{,}4\delta(f+10k) + 2{,}4\delta(f-9k).e^{j\pi/2} + 2{,}4\delta(f+9k).e^{-j\pi/2} + 2{,}4\delta(f-11k)e^{-j\pi/2} + 2{,}4\delta(f+11k)e^{j\pi/2}$


|H(f)|
12,4
2,4
9
10
11
f(kHz)


Θy(f)
π/2
f(kHz
-11
-10
-9
9
10
11
-π/2